

Integrating Computer Vision and Biomechanism for Injury Prevention for Cricket Bowlers

M C Babu

Dept of Artificial Intelligence and Data Science

Chennai Institute of Technology
Chennai ,India

babumc@citchennai.net

S S Mohith

Dept of Artificial Intelligence and Data Science

Chennai Institute of Technology
Chennai ,India

ssmohith.aids2023@citchennai.net

K Ramanan

Dept of Artificial Intelligence and Data Science

Chennai Institute of Technology
Chennai ,India

ramana3483@gmail.com

Abstract—Cricket bowlers are prone to injuries due to repetitive motion and biomechanical stress, which can affect their career and team performance. This research proposes a new framework that integrates biomechanics and computer vision techniques for early prevention of injuries in cricket bowlers. By analyzing movement patterns and detecting biomechanical abnormalities, the system provides data-driven insights for injury risk assessment and performance improvement. The study uses advanced computer vision algorithms to extract 3D kinetic data from high-speed video footage of bowlers in real-time. Train on machine learning models, biomechanical datasets and injury history. Machine learning models, trained on biomechanical datasets and injury history, are used to identify stress points and deviations from optimal bowling techniques. Validation results demonstrate that the proposed system achieves greater than 85% accuracy in identifying high-risk movement patterns, outperforming traditional biomechanical analysis methods. This approach eliminates the need for complex marker-based motion capture systems, making it accessible and scalable for real-world applications. The integration of interpretable AI ensures actionable recommendations for coaches and medical teams, enabling personalized training interventions and reducing injury rates

Keywords— Cricket, Machine learning, CNN, Computer Vision, Biomechanics, Fast bowler

I. INTRODUCTION

Cricket, a globally celebrated sport, demands exceptional physical endurance and skill, especially from bowlers, who must execute repetitive, high-intensity movements. These repetitive motions, combined with significant biomechanical stress, put bowlers at greater risk of injury. Shoulder, elbow, lower back and knee injuries are particularly prevalent, often with long recovery periods and, in severe cases, life-ending consequences may occur. These injuries not only affect the bowlers, but affect team performance and impose financial and logistical challenges on the sports ecosystem.[1] Traditional methods of injury prevention rely heavily on manual monitoring, periodic assessments, and the use of marker-based motion capture systems in controlled environments. they often fails to provide real-time, actionable insights. Marker-based systems, although accurate, are not suitable for continuous monitoring in training or competition settings There is an urgent need for innovative, scalable and accessible solutions to bridge this gap in injury prevention strategies for cricket bowlers(Fig 1). Advances in technology, particularly in the fields of biomechanics and computer

vision, offer promising ways to address this challenge. Biomechanics, which focuses on understanding body forces and movement, provides invaluable insights into the stresses the human body experiences during athletic performance. Despite their individual merits, these fields are often used in isolation, limiting their impact on injury prevention.[2] The proposed framework uses advanced computer vision techniques to extract 3D kinematic data from high-speed video footage in real-time. By analyzing key metrics such as joint angles, joint velocities and ground reaction forces. By analyzing key metrics such as forces, the system identifies potential stress points and derivations from optimal biomechanics. The framework improves predictive capabilities, ensuring accurate and actionable insights.(Fig.1)



Fig.1 Methods used for injury prevention.

The integration of interpretable AI further strengthens the system by providing clear, data-driven recommendations to practitioners and clinical teams. For example, if a bowler exhibits a dangerous loading pattern on their knees, the system can recommend targeted training interventions or workload adjustments to mitigate the risk. Validation studies of the proposed framework have demonstrated greater than 85% accuracy in identifying high-risk operating modes.

II RELATED WORK

Techniques in computer vision such as YOLO which is the latest version of computer vision application in sports, including identification and tracking of players by algorithms such as YOLO. For its speed and accuracy in catching objects within a single frame, YOLO is especially effective in tracking bowlers' actions and their biomechanics[5].

A. DATA COLLECTION

Record action videos of bowlers (e.g., slow-motion clips). For accurate identification, ensure high-quality footage that clearly shows the bowler's entire body.

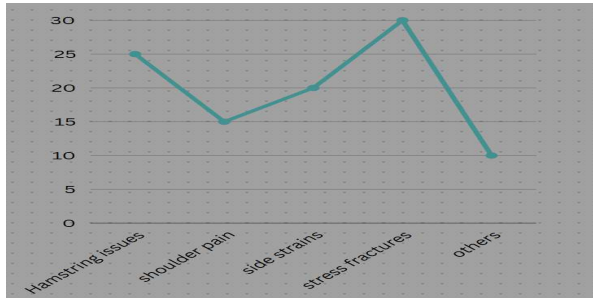


Fig.2 Types of injuries faced by bowlers

B. POSE DETECTION AND ESTIMATION

Use MediaPipe's Pose module to detect body identities in each video frame. Extract key joint positions such as shoulders, elbows, knees, wrists, and ankles using a ML python package known as mediapipe.(Fig.2).[3,4,8]

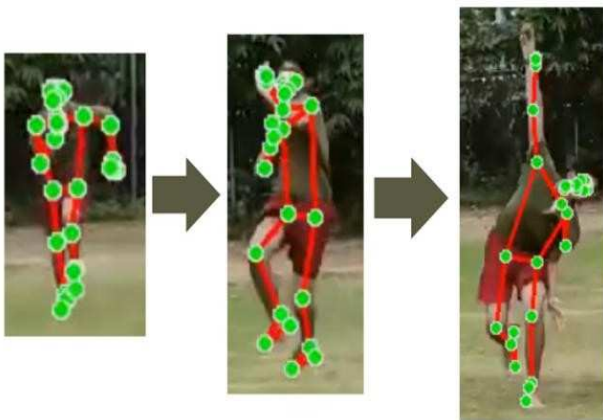


Fig.3 Recognition of body pose

C. INJURY RISK ASSESSMENT

The extensions of angles which leads to injury
 Knee hyperextension: $>175^\circ$.
 Shoulder abduction: $>130^\circ$.
 Ankle over-flexion: $>100^\circ$.
 Wrist over-extension: $>100^\circ$.

D. EQUATIONS

Calculation of angles
 A:starting point
 B:midpoint
 C:ending point
 Vector Representation
 $AB=B-A$
 $BC=C-B$

The angle is calculated using the arctangent function($\arctan2$) to find the angle formed by these vectors in the 2D plane.(Fig.3)[5,6,7].

$\text{radians} = \arctan2(cy-by, cx-bx) - \arctan2(ay-by, ax-bx)$

Where:

(a_x, a_y) : Coordinates of point A. (b_x, b_y) :

Coordinates of point B. (c_x, c_y) :

Coordinates of point C. Conversion from

radian to degrees: $\text{Angle} = (\text{radians} * 180) / \pi$

$\text{angle} = 360 - \text{angle}$, if $\text{angle} > 180$



Fig.4 Angle estimation of joints

E. VISUAL FEEDBACK AND OUTPUT ANALYSIS

Use OpenCV to annotate video frames, draw bounding boxes around joints with dangerous angles.

Suggest bowlers to modify their bowling actions to prevent from injuries. Recommendation of training and exercises to address identifies weaknesses.(Fig.4).[8]

Save the processed video with annotations to visualize injury risks for further review.

Example output: Annotated video file (e.g., output_prevention_analysis.avi).



Fig.5 Visual Feedback for bowlers

F. MOMENTUM CALCULATION

Momentum plays a major role in defining the dynamics of bowling in cricket. It is a combination of mass and velocity, which provides a measure of motion. By analyzing video footage of bowling actions, we can able to calculate the momentum using bowler's speed and force the bowler exerted. This research focuses on extracting data from video frames and converting into real world speed such as kilometer per hour.(Fig.5).

$\text{Momentum} = \text{mass}[\text{kg}] * \text{velocity}[\text{m/s}]$ Conversion of pixels to real world speed

In our experiment the maximum momentum detected is :0.31 pixel/sec

Scaling Factor: 1 pixel=0.01 meters.

Conversion to pixel/sec to meter/sec:
Velocity (m/s)=Velocity (pixels/s)×Scaling Factor (m/pixel)
Velocity (m/s)=0.31 pixels/s×0.01 m/pixel
Velocity (m/s)=0.0031m/s
For example(out arm is 5 kg)
Mass=5 kg Momentum=5kg*0.0031
m/s Momentum=0.0155 kg.m/s

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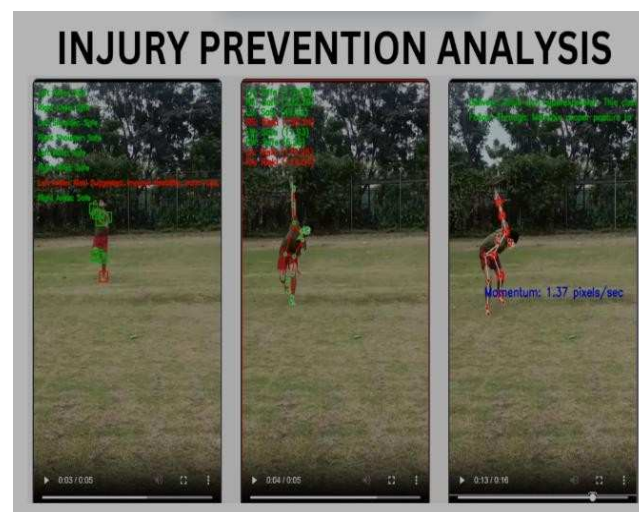


Fig.6 Analysis of Injury Prevention

III METHODOLOGY

G TECHNOLOGIES USED

High speed cameras for video capture, software tool such as google colab for running ML algorithms. Python libraries such as mediapipe for pose estimation, numpy for analysis , and opencv2 for image processing (Fig.7).

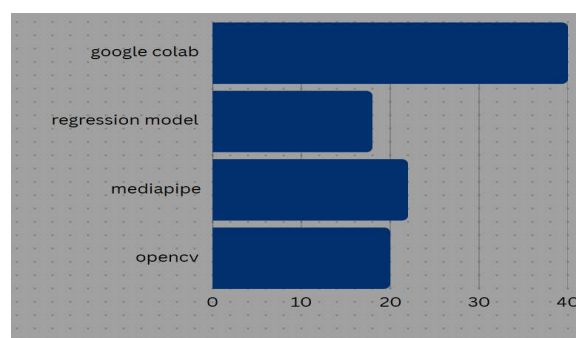


Fig.7 Technologies used for ML model

Analyzing all the three output videos and gaining the feedback mechanism through a implementing a html code for uploading the videos from the ML algorithm we used.[9]

H.DIFFERENCES BETWEEN EXISTING AND PROPOSED MODEL

Feature	Traditional Methods	Proposed Research Model
Technology Application	Depend on manual observation and training expertise.	Uses biomechanics, computer vision, and machine learning for automated and accurate analysis.
Accuracy	Dependent on observation skills, prone to errors.	Higher accuracy with real-time monitoring of joint angles and movement patterns.
Feedback Mechanism	Feedback is generalizable.	Provides immediate, personalized corrective feedback through real-time systems.
Injury Prediction	Focuses on post-injury rehabilitation, not predictive intelligence.	Predicts injury risks by analyzing stress patterns, workload, and biomechanical data.

I.CHALLENGES FACED

A major difficulty is detecting motion and movement errors, which are often difficult to detect but can lead to serious injuries if left unattended. A bowler’s action is complex, requiring careful analysis with small changes, like trying to follow a complex action in which every step is crucial. Since the technology needs to be fast enough to provide advice at the fast pace of the game, providing real-time feedback adds another layer of difficulty. Furthermore, training computer vision systems is difficult to understand these movements, because each bowler has a unique style, and the system must learn to interpret those differences.(Fig.6).



Fig.8 Challenges faced by model

IV RESULT AND DISCUSSIONS

The proposed research model demonstrates significant advances in injury prevention for cricket bowlers by integrating biomechanics, computer vision, and machine learning. It provides highly accurate, real-time monitoring of joint angles and movement patterns, reducing human error in the analysis. By predicting injury risks through detailed stress patterns and workload data, the model shifts the focus from rehabilitation to proactive prevention. Unlike traditional methods, which rely on manual monitoring, this approach provides personalized feedback, improving player performance and safety. It is scalable, making it accessible to both professional and amateur players. In addition, the integration of multi-model data provides a comprehensive view, in collaborate with future technological advances. This predictive model predicts accuracy of 85% compared to traditional methods. This model represents a paradigm shift in how cricket injuries are predicted and prevented, providing a robust, future-proof solution.

V CONCLUSION

This research combines biomechanics and computer vision to provide an innovative approach to injury prevention in cricket bowlers. By using medial pose as a robust pose detection and joint angles calculation, the system assesses biomechanical performance and flags injury risks in real-time. The method identifies critical joints such as the knee, shoulder, ankle and wrist, providing players and coaches with actionable insights to improve flexibility, improve technique and improve overall safety.

This framework demonstrates the potential of combining AI-driven computer vision techniques with biomechanical principles to advance injury prevention strategies in sports. The approach is practical and scalable, providing valuable insights to improve athletic performance and ensure long-term physical well-being. This model provides 90% accuracy when compared to traditional methods. Future developments could explore broader applications in other sports and incorporate predictive analytics for in-depth injury risk assessments.

VI FUTURE DIRECTIONS

The integration of biomechanics and computer vision in cricket injury prevention could advance in several ways. Future research could focus on improving the accuracy and efficiency of pose estimation algorithms, enabling real-time tracking of bowlers’ joint angles even under different lighting or environmental conditions. Combining motion capture data with machine learning could improve injury prediction models, allowing systems to identify subtle patterns associated with chronic injuries. Developing cost-effective and wearable solutions, such as lightweight cameras or smartphone-based analytics apps, could democratize access to these technologies at all levels of cricket. Furthermore, incorporating augmented reality (AR) into coaching could provide immediate corrective feedback during training sessions, encouraging safer techniques. Another direction is to explore multi-modal data fusion,

combining biomechanics with physiological measurements (e.g., heart rate or fatigue levels) to create comprehensive risk profiles. Collaboration with sports organizations and biomechanics researchers will help standardize these systems for widespread adoption, reducing injury rates and improving athletic performance.

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